EXCERCISE 2 SQL AGGREGATE & Operat

01 STUDENT TABLE

1) SELECT DISTINCT Department
FROM students;

DEPARTMENT
IT
HR
Finane

2) SELECT department,
AVG (Age) AS avg_age
FROM student
GROUP BY department;

DEPARTMENT	Avg_age
IT	20,5
HR	22
Finance	23

(3)
```
SELECT department, COUNT(*) AS student_count
FROM students
GROUP BY department
HAVING COUNT(*) > 1
```

Department	Student
IT	2
HR	2

(4)
```
SELECT ~~id,~~ student id,
       name,
       age,
       department
FROM ~~dep~~ students
WHERE age BETWEEN 21 AND 23;
```

Student ID	Name	Age	DEPARTMENT
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

(5)
```
SELECT student id,
       name,
       age,
       department
FROM students
WHERE department IN ('IT', 'HR')
  AND age > 21;
```

Student ID	Name	Age	DEPARTMENT
2	Bob	22	HR
5	Eve	22	HR

② THE COURSES TABLE

06 SELECT department,
total_credits
FROM courses
GROUP BY department
HAVING SUM (credits) > 5;

Department	Total-credit
IT	11

7 SELECT course_id,
course_name
credits
FROM cources
WHERE ~~GROUP BY~~ (credits) < 4;

Course ID	Course name	~~Course~~ credits
101	SQL Basics	~~17~~ 3
104	Excel	2
105	statistics	3

(8) SELECT course_id,
Coure_name,
credits
FROM courses
ORDER BY credits DESC
LIMIT 3;

Course ID	Course Na	Credits
102	~~Phy~~ Python	4
103	DATA Scienc	4
101	SQL Basics	3

(03) THE ENROLLMENTS TABLE

(9) SELECT max_grade,
min_grade,
avg-grade

FROM enrollments;

Max_grade	Min_grade	AVG_grade
90	78	84,6

(10) SELECT course_id,
COUNT (*) AS enrollment_count

FROM enrollments
GROUP BY course_id;

course_id	Enrollment_count
1	2
2	3

04) THE SALARIES TABLE

(11) SELECT department,
SUM (salary) AS total_salary,
SUM (bonus) AS total_bonus

FROM salaries
GROUP BY department

Department	Total salary	Total bonus
IT	122 000	~~132 500~~ 10 500
HR	595 000	~~602 500~~ 7 500
Finance	70 000	6000

(12)

```
SELECT department,
       avg_salary

FROM salaries
WHERE salary > 55 000
GROUP BY departments,
HAVING AVG (salary) > 55 000
```

Department	Avg_salary
IT	91 000
HR	82 000

(13)

```
SELECT employee_id,
       name,
       salary,
       bonus,
       total_compensation
FROM salaries
GROUP BY departments
WHERE (salary+bonus) > 60 000
```

Employee ID	name	salary	bonus	Total Compensation
1	Tom	60000	5000	65 000
~~2~~	~~Jerry~~	~~55000~~	~~4000~~	~~59 000~~
3	Spike	76 000	6000	76 000
4	Tyke	62 000	5500	67 500

(14)

```
SELECT department,
       total_budget,
       avg_budget
FROM projects
GROUP BY department
HAVING AVG (budget) > 70000;
```

Department	total_budget	avg_budget
IT	270 000	135 000
Finance	80 000	80 000

(15)

```
SELECT project_id,
       project_name
       department
       budget
FROM projects
WHERE budget BETWEEN 50 000 AND 120 000
      AND department = 'Marketing';
```

Project_id	project_nam	departmen	budget
1	AI App	IT	120 000
2	Payroll System	Finance	80 000
3	Dashboard	IT	150 000
5	HR Portal	HR	50 000